CITS5508: Assignment 2

Semester 1, 2026

Assessed, worth **20%**. Due **11:59 pm Friday 15th May 2026**.

1 Outline

This assignment is divided into two parts: an investigation of *Ensemble Learning* methods on the *Heart Disease dataset*, and an unsupervised investigation combining *Dimensionality Reduction* and *k-Means clustering* on the *Olivetti Faces dataset*. Detailed descriptions of each task are given below.

While this assignment is centred around programming, you should treat your submission as a report. A specific set of tasks is given for both **Part 1** and **Part 2**; however, your assignment is to perform an investigation. Do not expect to receive full marks for one-word answers when asked to comment on your results. Explain your thoughts in detail.

2 Submission

- You should submit your assignment via LMS. The submission should be a single Jupyter Notebook file (with an `.ipynb` extension).
- Your notebook should include all code, outputs, descriptions and comments. Treat it as a report – introduce the topic, describe your methodology, present your results and conclusions.
  - Make use of Markdown cells inserted at appropriate places to explain your code. In addition to Markdown cells, some short comments can be put alongside the Python code
  - Dividing the assignment into suitable sections and subsections (with section and subsection numbers and meaningful headings) would make your portfolio easier to follow.
  - Please see `sample.ipynb` under the Week 2 content on LMS for an example of how to structure your notebook.
- Your notebook should already include all output, but must also run without errors when using the CITS5508 Anaconda environment. Note that Google Colaboratory will produce identical outputs to our environment.

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3 Part 1 – Ensemble Learning for Heart Disease Prediction

Your first task is to compare a range of **ensemble methods** for predicting the presence of heart disease in patients.

You will use the *Heart Disease dataset* from the UC Irvine Machine Learning Repository, available at:

https://doi.org/10.24432/C52P4X

Download the dataset from LMS and place the extracted files in the same directory as your notebook. You may wish to take a look at the UCI website for more information about the dataset, but do not use the "import in Python" option.

Please use the `processed.cleveland.data` file from LMS. Do not use the "import in Python" option from the website.

3.1 Tasks

1. Load the dataset and assign appropriate column names. Briefly describe the dataset:
   (a) How many examples are there? What does each example represent?
   (b) How many features are there? Which are numeric and which are categorical?
   (c) Describe the target attribute (`num`). The original target indicates the severity of heart disease on an integer scale from 0 to 4. For this assignment, **binarise the target**: 0 indicates absence of disease, and any value greater than 0 indicates presence.
2. Split the dataset into a training and testing set. Use 80% of the data for training, and 20% for testing. Make sure to stratify your split.
3. Fit a single Decision Tree classifier using scikit-learn as a baseline and evaluate its performance on the test set.
4. Fit two further ensemble models using scikit-learn:
   - A Random Forest classifier with 500 estimators.
   - An AdaBoost classifier with 500 estimators.
5. Combine your models from the previous steps in a soft-voting **Voting Classifier**.
6. Compare the performance of all four models (the decision tree baseline, the Random Forest, the AdaBoost classifier, and the voting classifier) on the test set:
   (a) Create a table of precision, recall, and F1-scores for each model.
   (b) Display a confusion matrix for each model.
7. Plot the feature importances of your Random Forest. Briefly comment on which features contribute most strongly to predictions, and whether this matches what you would expect.
8. Comment on your results in detail. Investigate the following ideas:
   (a) How do the models compare to each other? Which model performs best? Is this what you expected?
   (b) Bagging and boosting reduce different sources of error. Which type of ensemble works better here, and why might that be?
   (c) What are the advantages and disadvantages of using an ensemble method over the decision tree baseline?

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4 Part 2 – Clustering of Faces

Your second task is an unsupervised investigation of the *Olivetti Faces dataset* from AT&T Laboratories Cambridge, combining **Principal Component Analysis (PCA)** with k-Means clustering.

You can load the dataset directly from scikit-learn using the following code:

```
from sklearn.datasets import fetch_olivetti_faces
data, targets = fetch_olivetti_faces(return_X_y=True)
```

Figure 1: Example images from the dataset.

4.1 Tasks

1. Load the Olivetti Faces dataset using the code above. Briefly describe the dataset:
   (a) How many examples are there? What does each example represent?
   (b) Describe the features. How many features are there, and what does each feature represent?
2. Sample a random example from each class and display the images in a grid. Ensure each image is labelled with its class.
3. Fit Principal Component Analysis (PCA) on the full dataset using scikit-learn.
   (a) Plot the cumulative explained variance ratio against the number of components. Choose a number of components that retains at least 95% of the variance and report this number.
   (b) Display the first 10 principal components as 64×64 images. These are known as *eigenfaces*. Explain what these components represent and how they relate to the original images.
   (c) Project the data onto the first two principal components and produce a scatter plot, colouring each point by its class. Comment on whether 2 components are sufficient to separate the classes.
4. Apply k-Means clustering to the original feature space. Use k equal to the number of classes.
   (a) Report the silhouette score of the clustering.
   (b) Visualise the cluster centroids as 64×64 images. Compare them to the eigenfaces from the previous task.
5. Repeat the k-Means clustering, this time on the PCA-reduced data.
   (a) Report the silhouette score for the PCA-reduced clustering.
   (b) Plot the silhouette diagrams for both the original and PCA-reduced clusterings, and compare them.
6. Comment on your results in detail. Investigate the following ideas:
   (a) How does PCA affect the quality of the k-Means clustering? Discuss the performance.
   (b) What are the advantages and disadvantages of using PCA before clustering, compared with clustering on the raw features?

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5 Marking Rubric

Part 1 – Ensemble Learning for Heart Disease Prediction

Task	Failed (0%)	Novice (33%)	Competent (66%)	Proficient (100%)
P1.1				
(2 marks)	Not attempted or incomplete.	Data not loaded correctly, or description contained errors.	Data loaded and described correctly, but lacking minor details.	Data loaded and described correctly with excellent detail.
P1.2				
(3 marks)	Not attempted or incomplete.	Data split incorrectly, with major issues.	Data split with minor issues only.	Data split correctly with appropriate stratification.
P1.3				
(5 marks)	Not attempted or incomplete.	Decision tree baseline fit incorrectly, with major issues.	Decision tree baseline fit correctly with minor issues only.	Decision tree baseline fit correctly with appropriate hyperparameters and evaluated cleanly.
P1.4				
(5 marks)	Not attempted or incomplete.	Random Forest and/or AdaBoost fit incorrectly, with major issues.	Both ensembles fit correctly with minor issues only.	Both ensembles fit correctly with appropriate hyperparameters.
P1.5				
(5 marks)	Not attempted or incomplete.	Voting classifier fit incorrectly, with major issues.	Voting classifier fit correctly with minor issues only.	Voting classifier fit correctly with appropriate hyperparameters.
P1.6				
(5 marks)	Not attempted or incomplete.	Inappropriate metrics, or major issues with calculation or presentation.	Appropriate metrics correctly calculated, but presentation has minor issues.	Appropriate metrics correctly calculated and well presented for all four models.
P1.7				
(5 marks)	Not attempted or incomplete.	Feature importance plot has major issues, or commentary is absent.	Feature importance plot correctly produced with brief commentary.	Feature importance plot correctly produced with thoughtful commentary.
P1.8				
(15 marks)	Not attempted or incomplete.	Brief analysis without a lot of exploration.	Good analysis, but lacking minor details.	Thorough analysis with excellent detail.

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Part 2 – Clustering of Faces

Task	Failed (0%)	Novice (33%)	Competent (66%)	Proficient (100%)
P2.1				
(2 marks)	Not attempted or incomplete.	Dataset description contained errors.	Dataset described correctly, but lacking minor details.	Dataset described correctly with excellent detail.
P2.2				
(3 marks)	Not attempted or incomplete.	Images not sampled or displayed correctly.	Images sampled and displayed correctly, but code was messy or hard to read.	Images sampled and displayed correctly. Code was not messy and ran successfully.
P2.3				
(15 marks)	Not attempted or incomplete.	PCA performed incorrectly, with major issues. Eigenfaces and scatter plot have major errors.	PCA performed correctly with minor issues only. Good analysis, but lacking minor details.	PCA performed correctly. Variance threshold well justified, eigenfaces interpreted insightfully, 2D scatter discussed clearly.
P2.4				
(10 marks)	Not attempted or incomplete.	k-Means fit incorrectly. Silhouette score or centroid plots have major issues.	k-Means fit correctly with minor issues only. Silhouette score and centroid plots presented with minor issues.	k-Means fit correctly. Silhouette score reported clearly. Insightful comparison of centroids to eigenfaces.
P2.5				
(10 marks)	Not attempted or incomplete.	k-Means on PCA-reduced data fit incorrectly, or silhouette diagrams missing or incorrect.	k-Means on PCA-reduced data fit correctly with minor issues only. Silhouette diagrams produced with minor issues.	k-Means correctly fit on PCA-reduced data. Silhouette score reported clearly. Silhouette diagrams produced and compared insightfully.
P2.6				
(15 marks)	Not attempted or incomplete.	Brief analysis without a lot of exploration.	Good analysis, but lacking minor details.	Thorough analysis with excellent detail across both sub-questions.

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